

Viktor Stanchev

Software Engineer

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<http://viktorstanchev.com>

Work experience

Anchorage Digital - Founding engineer

Jan 2018 - Present

- Co-inventor and implementer of a novel digital asset custody system, U.S. Patent No. 12,614,158 (granted Apr 2026)
- Designer and lead developer of Anchorage's self-custody product: Porto and of Atlas, their settlement network product
- Designer and lead of the open source VisualSign protocol for blockchain transaction visualization
- Technologies: Go, Kubernetes, Google Cloud, PostgreSQL, Temporal, Hardware Security Modules, etc.

Docker - Software Engineer

June 2015 - Dec 2017

- Developed Docker's highly available enterprise Docker artifact registry product
- Debugged and reported many low level docker, kernel and networking bugs
- Improved dev processes, mentored peers, architected features and solved difficult bugs
- Technologies: Go, Docker, etcd, RethinkDB, python, Jenkins, etc.

Square - Software Engineer

Sept - Dec 2014

- Automated vulnerability scanning and handling of newly discovered vulnerabilities
- Improved intrusion detection systems
- Discovered and fixed vulnerabilities
- Designed and built a simple system for issuing and managing TLS certificates
- Technologies: Ruby, Rails, SQLite, Nessus, OS X, Linux, etc.

Mozilla - Software Engineering Intern

Jan - Apr 2014

- Improved (performance, features) the Gecko Profiler and Cleopatra (frontend to the profiler)
- Added profiling capabilities to Talos (an automated performance testing framework)
- Technologies: C++, FirefoxOS, Python, JavaScript, HTML, CSS, etc.

LinkedIn - Software Engineer

May - Aug 2013

- Designed and developed a distributed system for monitoring the performance of LinkedIn's graph database (which gets on the order of 100,000 qps)
- Designed and developed a front-end for displaying metrics collected with the new system
- Built a foosball / ping pong ladder which uses PageRank to rank players (as a hackathon project)
- Technologies: Java, Scala, gradle, Norbert, JavaScript, node.js, Play framework, backbone.js, d3.js, etc.

LinkedIn - Software Engineer

Sept - Dec 2012

- Developed a system for monitoring front-end performance throughout all of linkedin.com
- Worked at all levels: front-end JavaScript to Hadoop data analytics and data visualization

- Technologies: Java, JavaScript, Hadoop, Kafka, Teradata, Node.js, SQL, dustjs, InGraphs, Red Hat Linux, etc.

Jan - Apr 2012

Tagged (renamed to if(we)) - Software Engineer

- Implemented a machine learning algorithm for spam detection
- Source paper: *F. Qian, A. Pathak, Y.C. Hu, Z.M. Mao, and Y. Xie, "A case for unsupervised-learning-based spam filtering"*
- Technologies: Java, Scala, JavaScript, Spring, Ubuntu, etc.

May - Aug 2011

SpeechBobble (renamed to Socialabra) - Web Application Developer

- Built front-end features in a private social network
- Found and fixed half of the top 10 OWASP 2010 risks
- Technologies: Java, JavaScript, MySQL, FogBugz, JBoss, Tomcat, etc.

Projects

Open source Bitcoin exchange: <https://txbits.org>

Jan 2014 - Nov 2015

- Designed, built, deployed and maintained a Bitcoin exchange
- Technologies: Scala, Play!, [web stuff], postgresql, ansible, logstash, elasticsearch, kibana, bitcoind, litecoind, nginx, memcached, spiped, runit

Network Code Module

Sept 2013

- Built a kernel module for RT linux that implements Sebastian Fischmeister's language for real time network schedules
- Source paper: *Sebastian Fischmeister, Oleg Sokolsky and Insup Lee, "Network-Code Machine: Programmable Real-Time Communication Schedules"*
- Technologies: Arch Linux, C, Python, bash, HTML, Bootstrap, JavaScript, CSS

Education

Bachelor of Software Engineering

Sept 2009 - June 2015

- University of Waterloo, Waterloo, ON, Canada
- Cumulative Average ~85%

Professional memberships

IEEE - Senior Member

Interests

End-to-end encryption, P2P, operating systems, network security, man in the middle attacks, social networks, privacy, reverse engineering, internet of things

Things not listed above

- Programming languages descending, by skill / experience: Scala, JavaScript, Python, Ruby, C++, C, Java, PHP, C#, Haskell
- Other: Linux, Android, Git, SVN, Mercurial, various security tools